Week 1 Internship Progress Report

City Scale Retrofitting Analysis for Ahmedabad

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Contents

1 Introduction

This report summarizes the learning activities and concepts explored during Week 1 of the internship at SWA Consultancy under the project titled “City Scale Retrofitting Analysis for Ahmedabad”.

The primary focus during this week was to build a strong theoretical understanding of:

- Building thermal behavior
- Insulation systems
- Green building concepts
- Building energy efficiency
- Urban Building Energy Modeling (UBEM)
- City-scale retrofit analysis workflows

The week mainly involved attending mentor discussions, understanding project objectives, studying thermal engineering concepts, and exploring sustainable building frameworks and simulation platforms.

2 Project Orientation

During the initial phase of the internship, daily meetings were conducted with mentors and team members to understand the project workflow and expected outcomes.

The discussions mainly focused on:

- Understanding the overall project roadmap
- Introduction to retrofit analysis concepts
- Importance of building energy efficiency in urban environments
- Ahmedabad climate conditions and cooling demands
- Understanding how city-scale simulations are performed

The orientation helped in developing a clear understanding of how existing urban buildings can be analyzed and improved using retrofit strategies to reduce cooling load and improve energy performance.

3 Thermal Concepts and Heat Transfer

A major portion of the week was dedicated to understanding thermal engineering concepts related to buildings and insulation systems.

3.1 Modes of Heat Transfer

The three fundamental modes of heat transfer studied were:

- **Conduction** – Heat transfer through solid materials.
- **Convection** – Heat transfer through moving fluids such as air.
- **Radiation** – Heat transfer through electromagnetic waves.

Understanding these mechanisms is important because heat transfer directly affects indoor thermal comfort and cooling energy demand in buildings.

3.2 Thermal Properties

Several important thermal properties related to insulation and building performance were studied.

- **Thermal Conductivity** – Measures the ability of a material to conduct heat.
- **Thermal Conductance** – Represents heat transfer through a material considering its thickness.
- **Thermal Transmittance (U-value)** – Represents the overall heat transfer through a building component. Lower U-values indicate better insulation performance.
- **Thermal Resistance (R-value)** – Represents resistance to heat flow. Higher R-values indicate better insulation.
- **Surface Film Conductance** – Resistance offered by material surfaces to heat flow.
- **Heat Flux** – Rate of heat transfer per unit area.
- **Thermal Diffusivity** – Describes how quickly heat spreads through a material.

These concepts are important for evaluating building envelope performance and cooling load reduction.

3.3 Heat Capacity Concepts

The following heat storage concepts were also studied:

- Specific heat capacity
- Volumetric heat capacity

These properties help explain thermal mass behavior and thermal buffering in buildings.

3.4 Moisture and Vapour Properties

The effect of moisture on insulation systems was also studied.

The concepts explored include:

- Vapour resistance
- Vapour permeability
- Permeance
- Moisture transmission
- Thermal emissivity

It was understood that insulation systems must remain dry because moisture significantly increases thermal conductivity and reduces insulation performance.

4 Insulation Materials

Different insulation materials and their thermal behavior were explored.

4.1 Natural Insulation Materials

The natural and fibrous insulation materials studied include:

- Fiberglass
- Glass wool
- Mineral wool
- Cellulose

4.2 Synthetic Insulation Materials

Synthetic insulation systems explored include:

- Expanded Polystyrene (EPS)
- Extruded Polystyrene (XPS)
- Polyurethane Foam (PUF)
- Polyisocyanurate (PIR)

4.3 Key Understanding

The following important observations were understood:

- Insulation performance mainly depends on trapped air or gas pockets.
- Closed-cell insulation systems generally provide better thermal and moisture resistance.
- Density and void relationships strongly influence insulation effectiveness.
- Wet insulation materials lose thermal efficiency.

5 Green Building Concepts

Several sustainable building and energy-efficient design concepts were explored.

The major concepts studied include:

- Climate-responsive architecture
- Passive cooling strategies
- Daylighting
- Building energy efficiency
- Sustainable construction materials
- Low embodied-energy materials
- Resource conservation principles
- Building envelope optimization

The importance of reducing solar heat gain and improving thermal performance in hot climates such as Ahmedabad was understood.

Passive strategies such as insulation, shading, reflective surfaces, and efficient building envelopes play a major role in reducing cooling demand and energy consumption.

6 Green Building Standards and Rating Systems

Different green building standards and rating systems used in India were studied.

6.1 IGBC

The Indian Green Building Council (IGBC) framework was explored to understand:

- Sustainable building assessment
- Energy efficiency
- Water conservation
- Indoor environmental quality

6.2 GRIHA

The GRIHA rating system was studied as India's national green building framework focusing on:

- Sustainable development
- Energy performance
- Resource efficiency
- Environmental impact reduction

6.3 ECBC 2017

The Energy Conservation Building Code (ECBC 2017) was explored to understand:

- Building envelope standards
- HVAC efficiency requirements
- U-value and SHGC concepts
- Cooling load reduction techniques

The differences between IGBC, GRIHA, and ECBC frameworks were also discussed.

7 Introduction to CityBES

The CityBES platform was explored to gain an initial understanding of Urban Building Energy Modeling (UBEM).

CityBES is a city-scale building energy simulation platform used for:

- Building energy analysis
- Retrofit scenario analysis
- Cooling load estimation
- Urban energy simulations

The platform provides visualization of buildings and energy data at district and city scales.

7.1 Initial Understanding Developed

The following concepts were initially understood through CityBES exploration:

- Urban Building Energy Modeling (UBEM)
- Baseline simulations
- Retrofit analysis workflows
- Building archetypes
- Energy performance comparison

It was understood that CityBES converts GIS-based building datasets into digital building energy models for evaluating energy efficiency and retrofit strategies.

8 Future Learning Goals

The planned next learning objectives include:

- Understanding Ahmedabad building dataset preparation
- Learning QGIS workflows
- Studying GeoJSON dataset creation
- Understanding building footprint extraction
- Exploring retrofit implementation workflows
- Understanding detailed building energy simulation processes

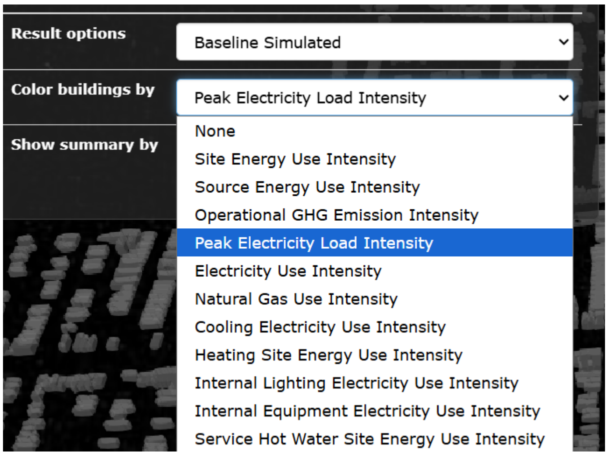

Figure 1: Different features explored

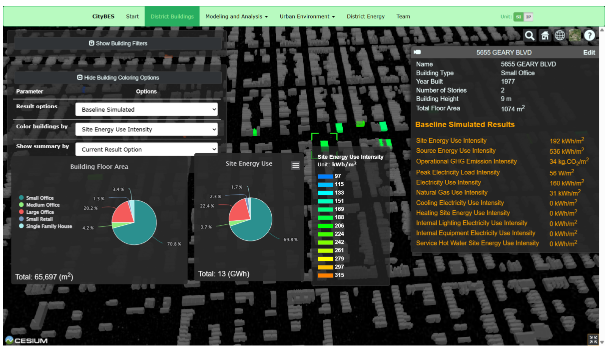

Figure 2: modules and results explored

9 Conclusion

Week 1 mainly focused on building theoretical foundations related to thermal engineering, insulation systems, green building concepts, and urban building energy analysis.

The concepts learned during this week provided a strong base for understanding:

- Building thermal behavior
- Energy-efficient building design
- Cooling load reduction
- Retrofit analysis
- Urban Building Energy Modeling workflows

The learning developed during this week will support future work related to Ahmedabad building datasets, GIS workflows, and CityBES-based retrofit simulations.